

CATS⁺

Competishum Advance

Training Sessions

(Lecture - )

Ques

The value of limit

$$\lim_{x \rightarrow 0^+}$$

$$\frac{e^{x^x-1} - x^x}{(x^{2x} - 1)^2}$$

is

☐ A $\frac{1}{2}$

☐ B $\frac{1}{4}$

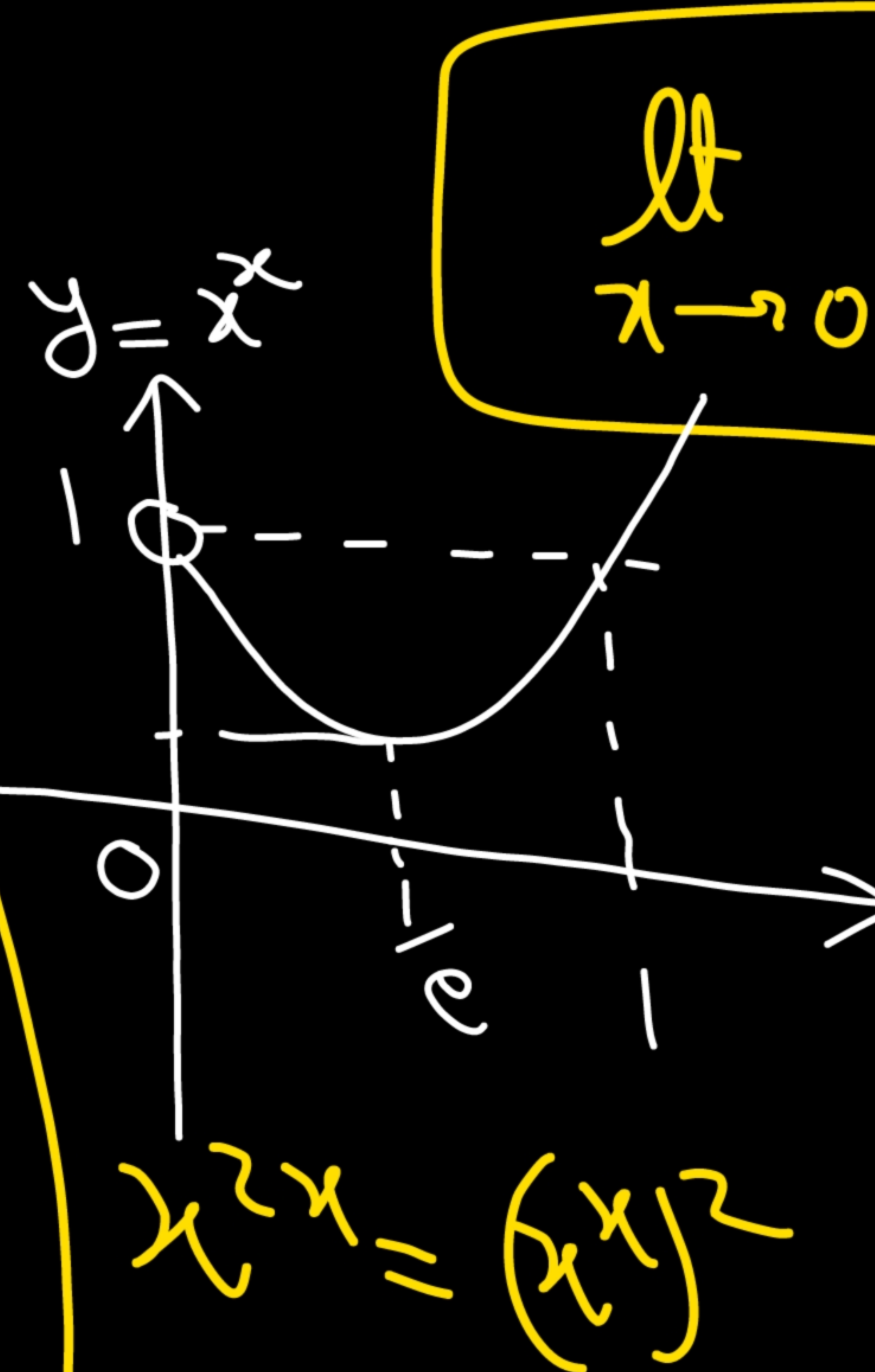
☒ C $\frac{1}{8}$

☐ D None of these.

Sol

$$\lim_{x \rightarrow 0^+} \frac{e^{x^x-1} - x^x}{(x^{2x} - 1)^2}$$

$$= \lim_{x \rightarrow 0^+} \frac{e^{x^x-1} - x^x}{\left((x^x-1)(x^x+1)\right)^2} = \lim_{x \rightarrow 0^+} \frac{e^{x^x-1} - x^x}{4(x^x-1)^2}$$



$$\lim_{x \rightarrow 0^+} x^x = 1$$

$$\begin{aligned} &= e^{\ln(x^x)} \\ &= e^{x \ln(x)} \\ &= e^{\frac{\ln(x)}{(\frac{1}{x})}} \end{aligned}$$

$$= \lim_{x \rightarrow 0^+} \frac{e^{x^x-1} - x^x}{(\underline{x^x-1})^2 \cdot 4}$$

$$= \lim_{t \rightarrow 0^-} \frac{e^t - 1 - t}{4 \cdot t^2}$$

$$= \lim_{t \rightarrow 0^-} \frac{\cancel{1+t} + \frac{t^2}{2} - \cancel{1-t}}{\cancel{4t^2}}$$

$$= \boxed{\frac{1}{8}} \text{ Am}$$

$$\begin{cases} x^x - 1 = t \\ \hookrightarrow x \rightarrow 0^+ \Rightarrow t \rightarrow 0^- \\ \Rightarrow x^x = 1 + t \end{cases}$$

Ques

If ' α ' is a root of the equation $\cos^{-1}x + \cos^{-1}2x + \cos^{-1}3x = \pi$, then $([.] = \text{GIF})$

(A) α is root of $12x^3 + 14x^2 - 1 = 0$

(C) $[\alpha] = 0$

(B) sum of possible values of α is $-\frac{7}{6}$

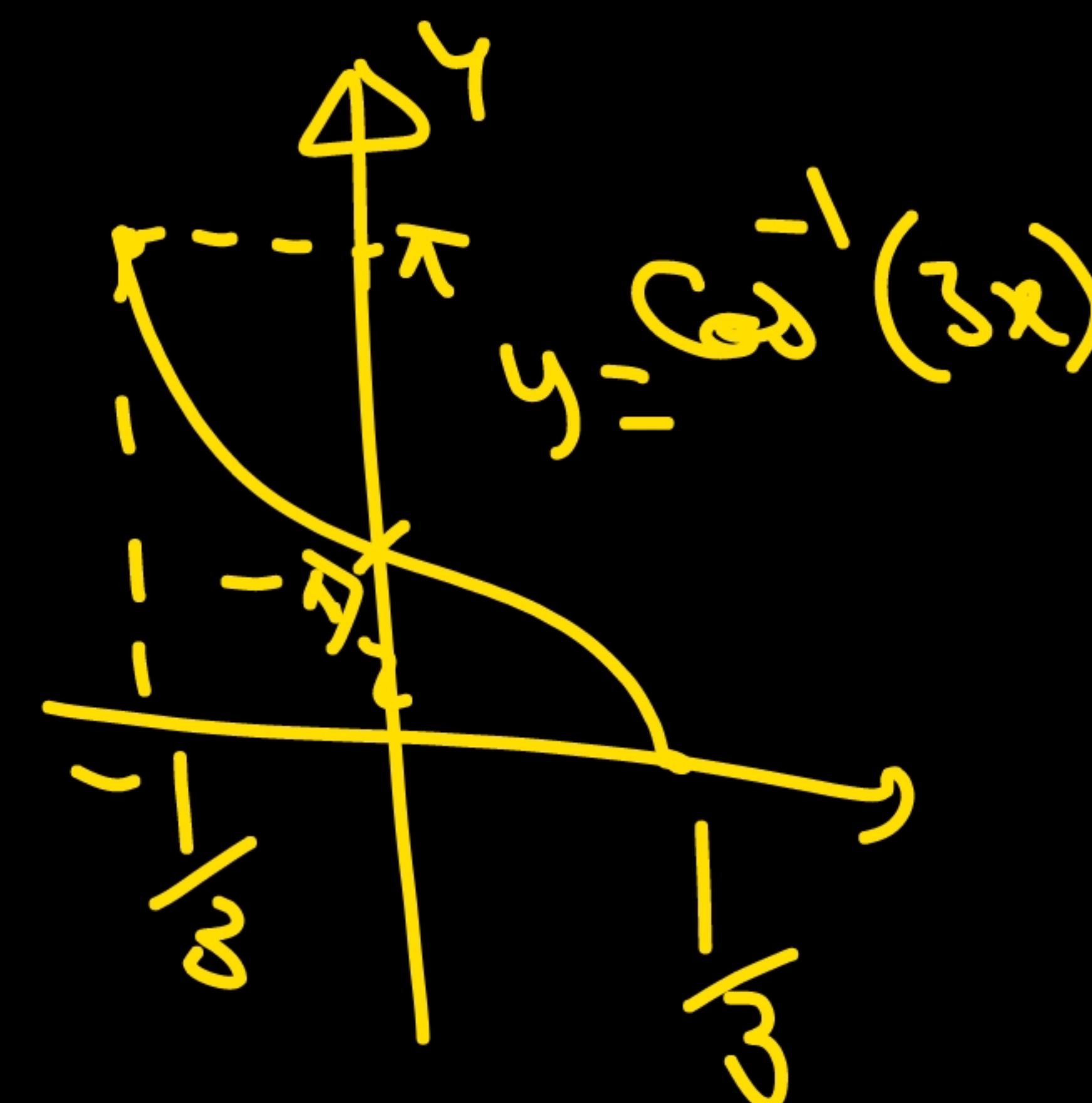
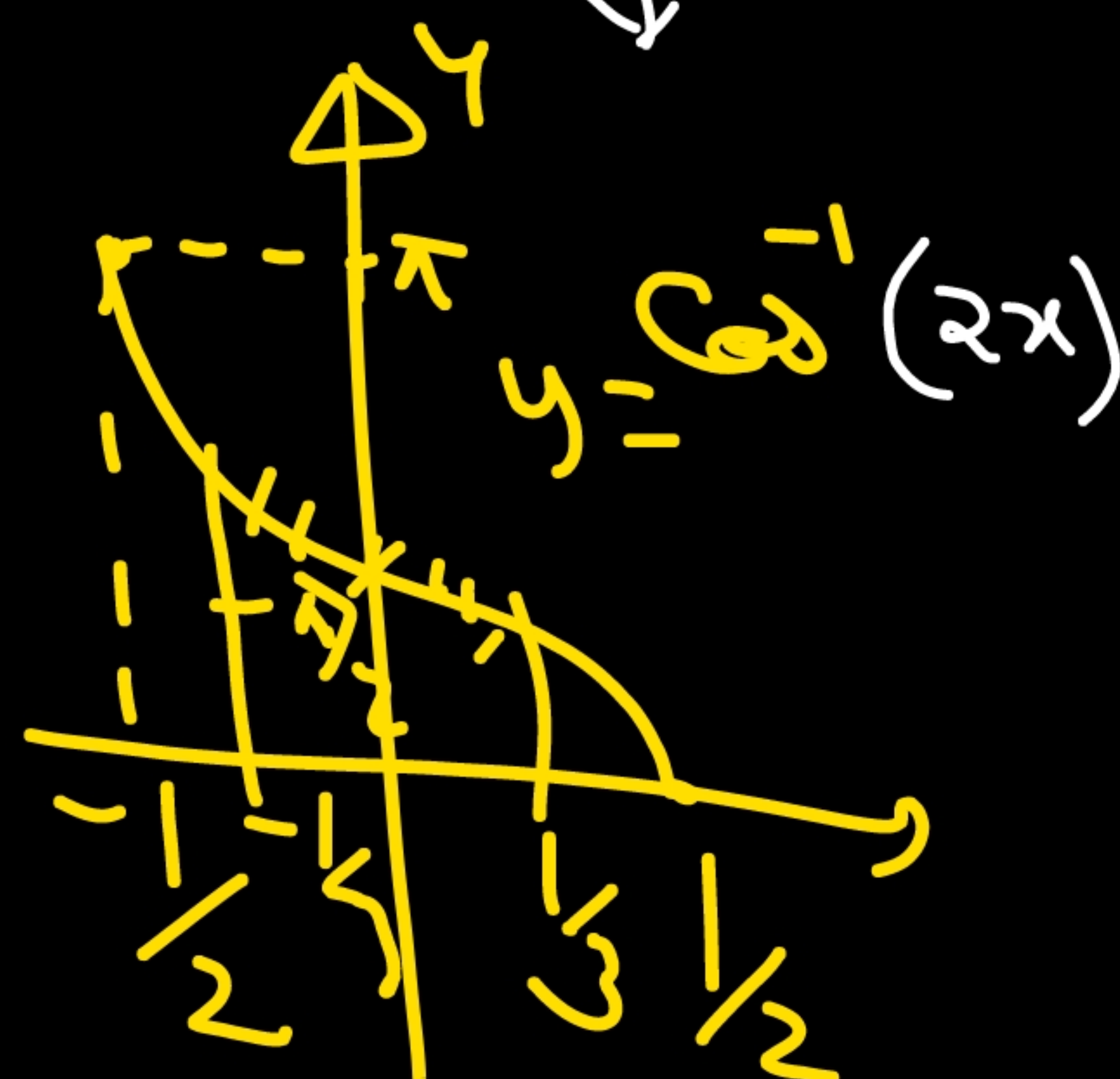
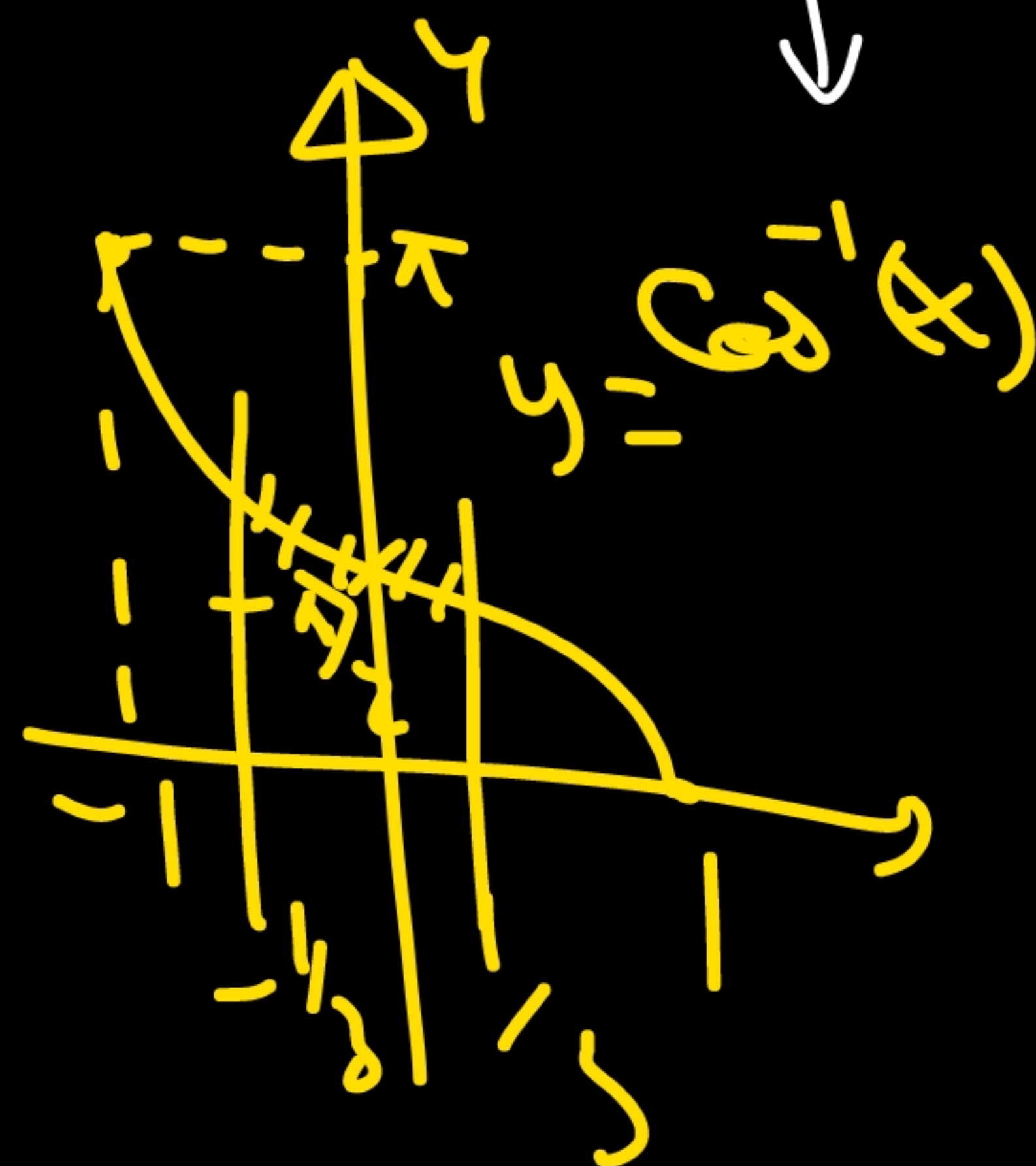
(D) three distinct real values of α are possible

Sol

$$\cos^{-1}(x) + \cos^{-1}(2x) + \cos^{-1}(3x) = \pi$$

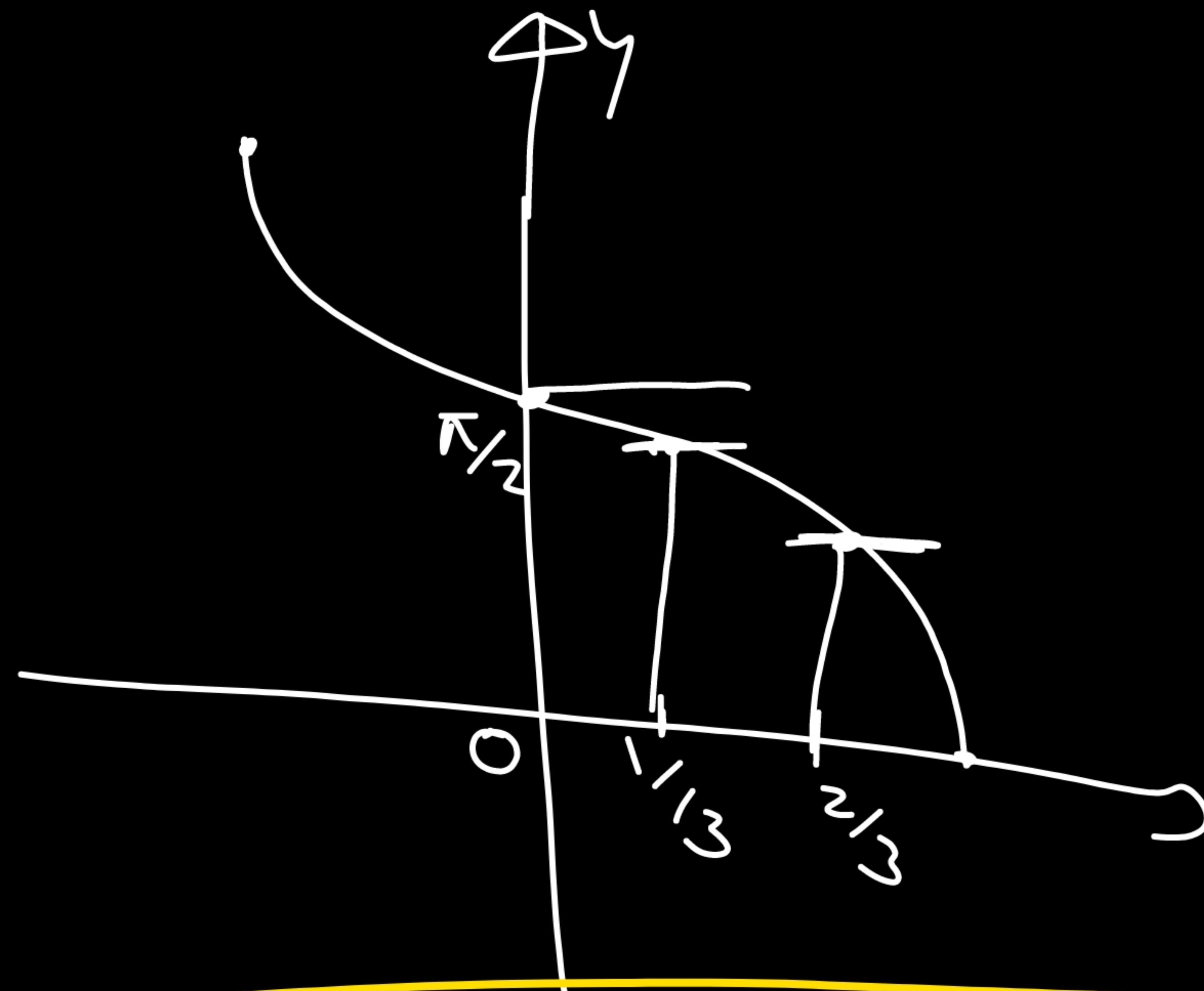
$$D_f = \left[-\frac{1}{3}, \frac{1}{3}\right]$$

$$\text{L.H.S.} = \cos^{-1}(x) + \cos^{-1}(2x) + \cos^{-1}(3x) \equiv \downarrow$$



$$f(x) = c^{-1}(x) + c^{-1}(2x) + c^{-1}(3x)$$

$$f\left(\frac{1}{3}\right) = \underbrace{c^{-1}\left(\frac{1}{3}\right)}_{< \frac{\pi}{2}} + \underbrace{c^{-1}\left(\frac{2}{3}\right)}_{< \frac{\pi}{2}} + \cancel{c^{-1}(1)}_0$$



$< \pi$

$$\cos^{-1}(x) + \cos^{-1}(2x) = \pi - \cos^{-1}(3x) = \cos^{-1}(-3x)$$

Case

$$x \cdot 2x - \sqrt{1-x^2} \sqrt{1-4x^2} = -3x$$

$$2x^2 + 3x = \sqrt{(1-x^2)(1-4x^2)} \xrightarrow{(\quad)^2}$$

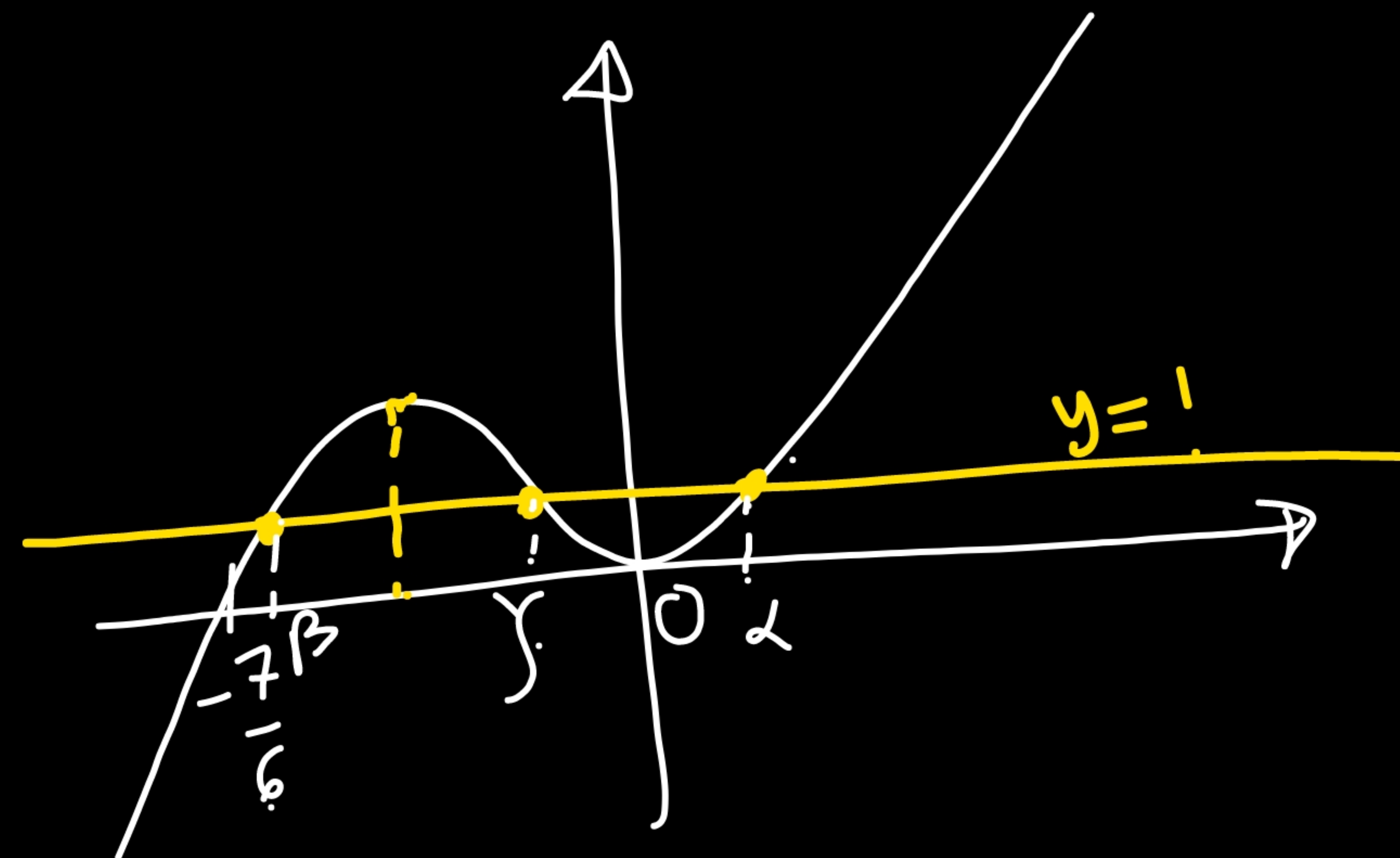
$$\cancel{4x^4} + 9x^2 + 12x^3 = \cancel{4x^4} - 5x^2 + 1$$

$$\boxed{12x^3 + 14x^2 - 1 = 0} \rightarrow \alpha, -$$

$$12x^3 + 14x^2 = 1$$

$$12\underline{x^2} \left(x + \frac{7}{6} \right) = \underline{1}$$

$\parallel$
 y



Ques

f is a twice differentiable function satisfying $f(0) = f'(0) = 1$. If $f''(x) \geq -3$ then the value of $f(4)$ can be

(A) -16

(B) -19

(C) -20

(D) -21

$$f(0) = f'(0) = 1$$

$$f(4) = ?$$

(A) (B)

$x > 0$

$$f''(x) \geq -3$$
$$\int_0^x f''(x) dx \geq -3 \int_0^x 1 \cdot dx$$
$$\Rightarrow [f'(x)]_0^x \geq -3(x-0)$$
$$\Rightarrow \underline{f'(x) - f'(0) \geq -3x} \Rightarrow$$

$$\Rightarrow \int_0^x f'(x) dx \geq \int_0^x (1-3x) \cdot dx$$
$$[f(x)]_0^x \geq \left(x - \frac{3x^2}{2}\right)_0^x$$
$$f(x) - \underline{f(0)} \geq x - \frac{3x^2}{2}$$
$$\boxed{f(x) \geq 1 + x - \frac{3x^2}{2}}$$

$$f(x) \geq 1 + x - \frac{3x^2}{2}$$

$$\downarrow x=4$$

$$f(4) \geq 1 + 4 - 24$$

$$f(4) \geq -\underline{\underline{19}}.$$

Que

If $f(x) = \sqrt{7x - x^2} - \sqrt{9x - x^2 - 14}$, then

(A) $f_{\min} = \sqrt{2}$

☒ (B) $f_{\max} = \sqrt{10}$

(C) $f_{\max} = 2\sqrt{5}$

☒ (D) $f_{\min} = 0$

Sol

$$f(x) = \sqrt{7x - x^2} - \sqrt{9x - x^2 - 14}$$

$$= \sqrt{\underset{+}{x}(\underset{+}{7-x})} - \sqrt{\underset{+}{(x-2)}(\underset{+}{7-x})}$$

$$= \sqrt{7-x} (\sqrt{x} - \sqrt{x-2})$$

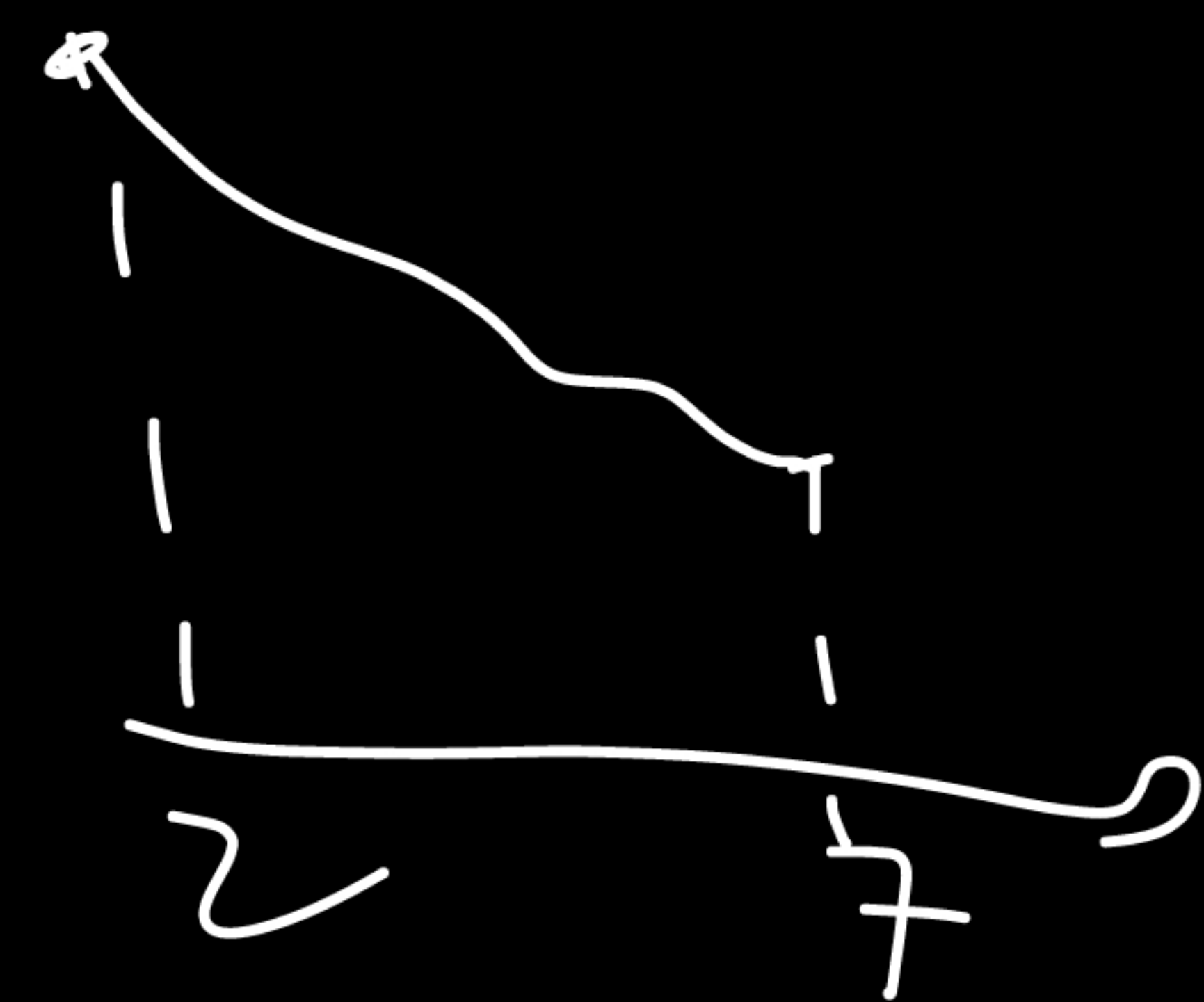
$$= \frac{2 \cdot \sqrt{7-x}}{\sqrt{x} + \sqrt{x-2}}$$

$\downarrow$
 $\uparrow$ $\parallel$ $\downarrow$ f_h

$$\Rightarrow f_{\max} = f(2) = \sqrt{10}$$

$$f_{\min} = f(7) = 0$$

$x \in [2, 7]$
domain



$$-(x^2 - 9x + 14)$$

$\downarrow$
 $7+2$ 7×2

BD

Ques

Four dice are rolled. Let A, B, C, D be the events that atleast one of the dice shows 3, 4, 5, 6 respectively. Then

(A) events A, B, C, D are equally likely

(C) $P(A \cap B) = 1 - 2 \left(\frac{5}{6}\right)^4 + \left(\frac{4}{6}\right)^4$

Sol 4-dice $\rightarrow$ rolled.

$$P(A) = P(\text{atleast one die shows 3})$$

$$P(B) = P(\text{atleast one die shows 4})$$

$$P(C) = P(\text{atleast one die shows 5})$$

$$P(D) = P(\text{atleast one die shows 6})$$

(B) $P(A) = 1 - \left(\frac{5}{6}\right)^4$

(D) $P(A \cap B | C \cap D) = \frac{12}{151}$

(B)

$$P(A) = 1 - P(\text{No die shows 3})$$

$$= 1 - \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} \cdot \frac{5}{6} = 1 - \left(\frac{5}{6}\right)^4$$

(C) $P(A \cap B) = 1 - P(\bar{A}) - P(\bar{B}) + P(\bar{A} \cap \bar{B})$

$$= 1 - \left(\frac{5}{6}\right)^4 - \left(\frac{5}{6}\right)^4 + \left(\frac{4}{6}\right)\left(\frac{4}{6}\right)\left(\frac{4}{6}\right)\left(\frac{4}{6}\right)$$

$$= 1 - 2 \cdot \left(\frac{5}{6}\right)^4 + \left(\frac{4}{6}\right)^4$$

$$\Rightarrow P(\bar{A}) = 1 - P(A) = \left(\frac{5}{6}\right)^4$$

$$P\left(\frac{A \cap B}{C \cap D}\right) = \frac{P(A \cap B \cap C \cap D)}{P(C \cap D)}$$

$$\begin{array}{r} 6^3 \times 3 \\ 216 \\ \underline{3} \\ 648 \\ 625 \\ \hline 23 \\ | 28 \\ \hline 151 \end{array}$$

$$= \frac{\left(\frac{24}{6^4}\right)}{1 - 2 \cdot \left(\frac{5}{6}\right)^4 + \left(\frac{4}{6}\right)^4}$$

$$= \frac{24}{6^4 - 2 \cdot 5^4 + 4^4}$$

$$= \frac{12}{648 - 625 + 128} = \frac{12}{151}$$

	D_1	D_2	D_3	D_4
$\left(\frac{1}{6}\right)^4$	4	3	6	5
	3	4	5	6
	3	4	6	5
	14			

Ques

Let a, b, c, d be any four real numbers. Then $a^n + b^n = c^n + d^n$ holds for any natural number n if
(A) $a + b = c + d$
(B) $a + b = c + d, a^2 + b^2 = c^2 + d^2$
(C) $a + b = c + d, a^3 + b^3 = c^3 + d^3$
(D) $a + b = c + d \neq 0, a^3 + b^3 = c^3 + d^3$

Soln $a, b, c, d \in \mathbb{R}$

$$a^n + b^n = c^n + d^n$$

~~(A)~~

$$\begin{array}{l|l} a+b=c+d & \\ \hline \hookrightarrow a=4 & c=5 \\ b=-4 & d=-5 \end{array}$$

$$n=2 \rightarrow 4^2 + (-4)^2 = 5^2 + (-5)^2$$

~~(A)~~

P_1 is true
(B) $\left\{ \begin{array}{l} a+b=c+d \\ a^2+b^2=c^2+d^2 \end{array} \right.$
 $\rightarrow a^2+b^2+2ab=c^2+d^2+2cd$
 $ab=cd$

$P_n, P_{n-1}, ab=cd$
 $\& a+b=c+d$
 $\Rightarrow P_{n+1}$ is true.

Let

$$P_n \equiv a^n + b^n = c^n + d^n \text{ is true.}$$

$$\begin{aligned} P_{n+1} &\equiv a^{n+1} + b^{n+1} = c^{n+1} + d^{n+1} \\ &= (a^n + b^n)(a+b) - ab^n - a^n b \\ &= (a+b)(a^n + b^n) - ab(a^{n-1} + b^{n-1}) \end{aligned}$$

$$(c+d)(c^n + d^n) - cd(c^{n-1} + d^{n-1})$$

① $\boxed{a+b=c+d} \neq 0$

$$a^3+b^3=c^3+d^3$$

$$\cancel{(a+b)}(a^2+b^2-ab) = \cancel{(c+d)}(c^2+d^2-cd)$$

$$\underline{a^2+b^2-ab} = \underline{c^2+d^2-cd}$$

$()^2$

$$\underline{a^2+b^2} + 2ab = \underline{c^2+d^2} + 2cd$$

$$\rightarrow 3ab = 3cd$$

$$\boxed{ab=cd}$$

✓

p_1, p_2

$$ab=cd$$

$$a+b=c+d$$

p_3

p_4

Ques If $x_i > 0$ for $1 \leq i \leq 6$ and $\sum_{i=1}^6 x_i = \pi$ then find the maximum value of $\sum_{i=1}^6 \sin x_i$

Ans 3

$$\frac{x_i > 0}{i=1, 2, 3, 4, 5, 6}$$

$$x_1 + x_2 + x_3 + \dots + x_6 = \pi$$

$$\hookrightarrow 0 < x_i < \pi$$

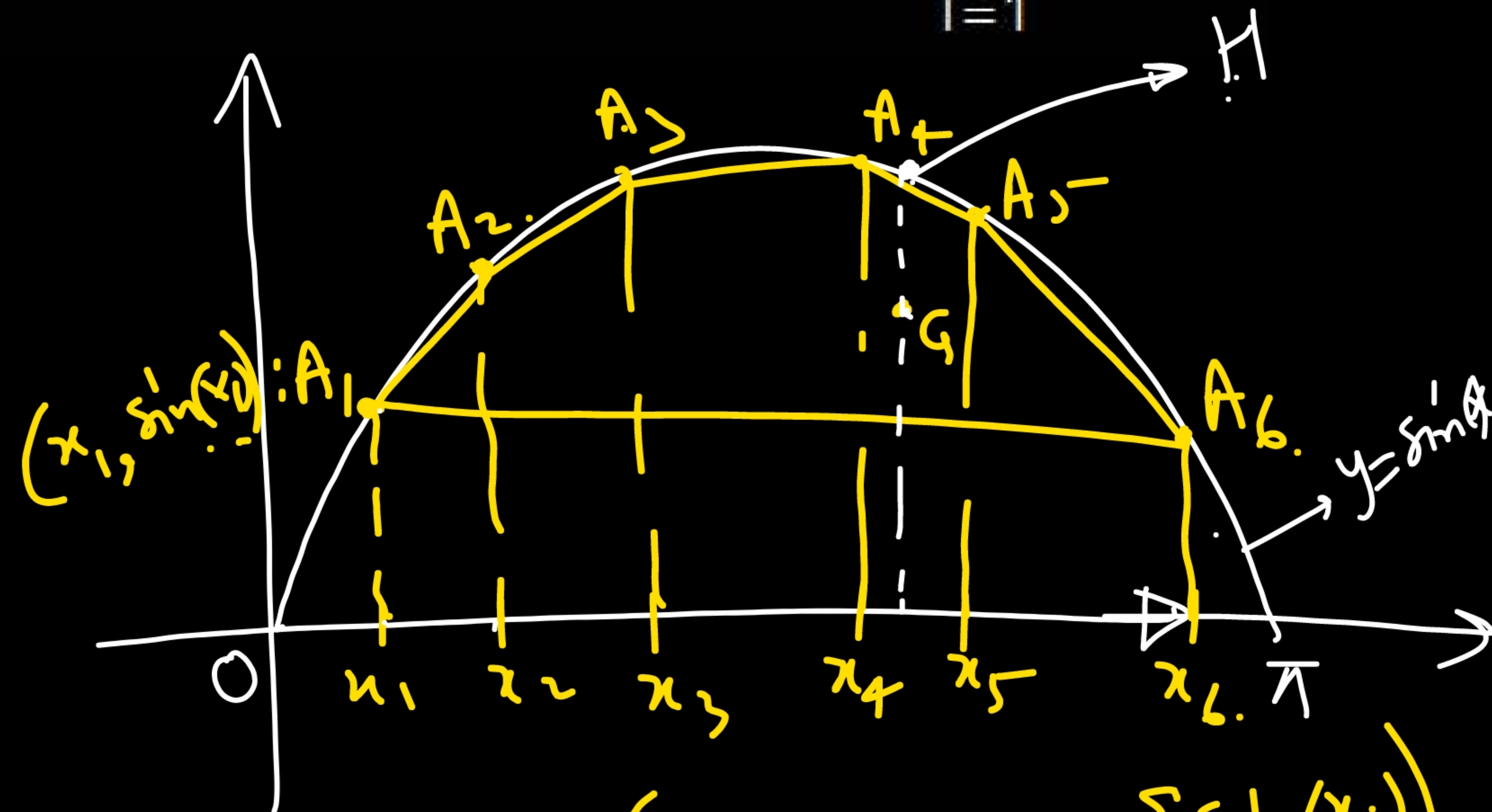
$$\sin(x_1) + \dots + \sin(x_6)$$

$$y_H \geq y_G$$

$$\sin\left(\frac{\sum x_i}{6}\right) \geq \frac{\sum \sin(x_i)}{6}$$

$$\sin\left(\frac{\pi}{6}\right) = \frac{1}{2}$$

$$\sum \sin(x_i) \leq 3$$



$$G: \left(\frac{x_1 + x_2 + \dots + x_6}{6}, \frac{\sum \sin(x_i)}{6} \right)$$

$$H: \left(\frac{\sum x_i}{6}, \sin\left(\frac{\sum x_i}{6}\right) \right)$$